BILE SALTS

&

BILE PIGMENTS

BILE

2. Bile is a bitter-tasting, dark green to yellowish brown fluid, produced by the liver.

Bile aids the process of digestion of lipids in the small intestine.

bile is stored in the gallbladder and upon eating is discharged into the duodenum.

Bile is a composition of the following materials: water (85%), bile salts (10%), pigments (3%),? fats (1%), inorganic salts (0.7%) and cholesterol (0.3)

COMPOSITION:

Daily secretion → 500 – 1000 ml Transparent alkaline fluid, light yellow in colour.
pH : 7.8 – 8.6
Water ,Bile Salts – These are sodium & potassium salts of bile acid.
Bile pigments –These are biliverdin & bilirubin.
5. lecithin, Fats, Fatty acids,Cholesterol, Enzyme Alkaline Phosphatase
Electrolytes

1. Anion – Cl^- , $Hco3^-$
2. 2. Cation – Na^+, K^+, Ca^{+2}

Bile Salts: These are synthesized in liver from cholesterol.

Primary bile acids – Cholic acid & Chenodeoxy cholic acid.

Conjugation – bile acids get conjugated in liver with Taurin & choline.

1. Cholic Acid + Taurine = Taurocholic Acid
2. Cholic Acid + Glycine = Glycocholic acid

Conjugated ‘Tauro’ and ‘Glyco’ cholic acid form sodium & Potassium Salt in bile.□ Sodium tauro cholate & potassium taurocholate.□ These are K/a Bile Salts.

90 – 95 % of Bile salts which enter in to duodenum are absorbed back in to Portal Vein.
Return back to liver.
This is called Entero Hepatic Circulation of Bile salts.
Remaining 5 – 10 %, enter the colon acted upon by colon Bacteria to form Deoxy cholic acid & lithocholic acid.
9. Total bile salts – 3.5 gms.
Recycle repeatedly via Entero Hepatic Circulation.
An ordinary meal needs 6 – 8 gm of bile salts to digest & absorb fats.
Total bile salts circulate twice during digestion of each meal.

Bile pigment: Bile pigments are bilirubin & biliverdin.

These are formed from globin part of Haemoglobin (Hb) after destruction of old RBCs.

No functional importance.

Excreted in bile. Responsible for yellow color of bile.

FUNCTIONS OF BILE

Bile acids are amphipathic, they contain both hydrophobic (lipid soluble) and polar (hydrophilic) faces.

Their amphipathic nature enables bile acids to carry out important functions:

Emulsification of lipid aggregates, Bile acids have detergent action on particles of dietary fat. Emulsification causes fat globules to break down or be emulsified into minute, microscopic droplets. Emulsification greatly increases the surface area of fat make it available for digestion by lipases,

CASEIN

Acetic Acid Test

- **OBJECT:** To detect the presence of casein in the given sample.
- **Principle:** Casein is precipitated out at its iso-electric pH(4.6) when 1% acetic acid is added drop wise..
- **Reagents:** 1% acetic acid, Bromocresol green indicator

Procedure:

- Take 5ml of given sample in a clean test tube.
- Add a drop of bromocresol green indicator.
- Add 1% acetic acid drop by drop till the color of solution changes to green (at pH 4.6).

Observation :

White precipitates appear.

Inference:

Casein is present.

Neumanns Test

- **OBJECT:** To perform the confirmatory test for casein in the given sample.

- **Principle:** On heating with conc. Sulphuric acid and nitric acid, casein is digested and phosphorus is released. Amonia solution is added to provide alkaline media. Ammonium molybedate react with phosphorus in alkaline medium and forms yellow precipitates of phospho ammonium molybedate.

- **Reagents:** conc. H_2SO_4, conc. HNO_3, Ammonium molybedate solution, Methyl red indicator, Ammonia solution.

Procedure:

- Take knifepoint of solid casein in a dry test tube.
- Add carefully 12 drops of conc. H_2SO_4 & 3 drops of conc. HNO_3.
- Heat gently until brown fumes changes to white.
- Then cool it.
- Add 3ml of H_2O followed by a drop of methyl red indicator.
- Add NH# solution till color changes to yellow.
- ADD 2ml of Ammonium molybedate solution and boil for 2-3 mins.

Observation :

Yellow precipitates of phospho ammonium molybedate appear.

Inference:

Casein is confirmed.

BILE SALTS

OBJECT: To detect the presence of bile salts in the given sample of urine by Hay's test

Principle: It is based on the fact that bile salt lowers the surface tension, allowing the sulfur powder to sink.

Reagents: sulfur powder, urine sample

Procedure: Take 2 test tubes & mark them as A &B & proceed as follows:

Procedure	Observation	Inference
In tube A: take 5ml of the given urine sample & sprinkle sulfur powder on the surface of solution.	Sulfur powder sink & settled down in the bottom of tube A (Bile salt lowers surface tension)	Bile salts are present in the urine sample
In tube B: take 5ml D.W & sprinkle sulfur powder on the surface of water	Sulfur powder remains on the surface of water in tube B	It is control in tube B

Result: **In the given sample of urine Bile salts are present.**

BILE PIGMENTs

OBJECT: To detect Bile pigments in the given urine sample by Gmelin's test.

Principle: Conc.HNO_3 oxidizes bilirubin in to various colored compounds namely Biliverdin (Green), Bilicyanin (Blue), Bilifuscin(Red) and Choletelin (yellow).

Reagents: Urine sample , Conc.HNO_3

Procedure: Take 2 test tubes & mark them as A & B, & proceed as follows:

Procedure	Observation	Inference
In tube A: take 3ml conc.HNO_3& add 3ml urine sample along the side wall of the tube carefully, to form multicolored rings	Multicolored rings formed due to the formation of oxidative products of bilirubin	Bile pigments are present in the given sample of urine.
In tube B: Take 3ml Conc.HNO3 & add from the side walls of the tube 3ml D.W	Multicolored rings not formed .	It is control

RESULT: Bile pigments are present in the given urine sample.